

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

Lecture: CPR 22

**Title: Pentose Phosphate Pathway:
Significance in RBCs and WBCs**

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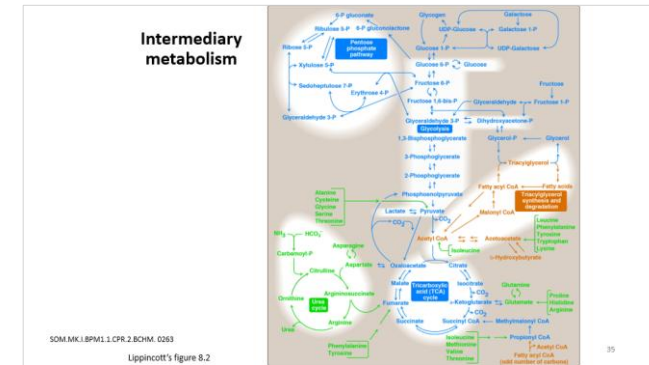
Lecture: Pentose Phosphate Pathway: Significance in RBC and WBC

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0262	List the tissues in which the PPP is active, and identify its subcellular location
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0263	Summarize the function of the PPP (metabolic roles of pentoses and NADPH).
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0264	Explain how malaria may be a factor in the relatively high rate of G6PD deficiency alleles in some populations
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0265	State the reactions of the oxidative portion of the PPP (emphasize the significance of G6PD)
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0266	Provide an overview of the intermediates of the non-oxidative phase of the PPP highlighting the importance of transketolase.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0267	Explain how the PPP is regulated and how different parts of the pathway are used, depending on the needs of the cell.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0268	Evaluate the use of NADPH in: Detoxification of hydrogen peroxide, Phagocytosis, Nitric oxide synthesis, Reductive biosynthesis
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0269	Describe glucose 6-phosphate dehydrogenase (G6PD) deficiency including precipitating factors, X-linked recessive inheritance, symptoms, laboratory findings, biochemical basis of findings (hemolysis and Heinz bodies in RBC).

Recommended Reading

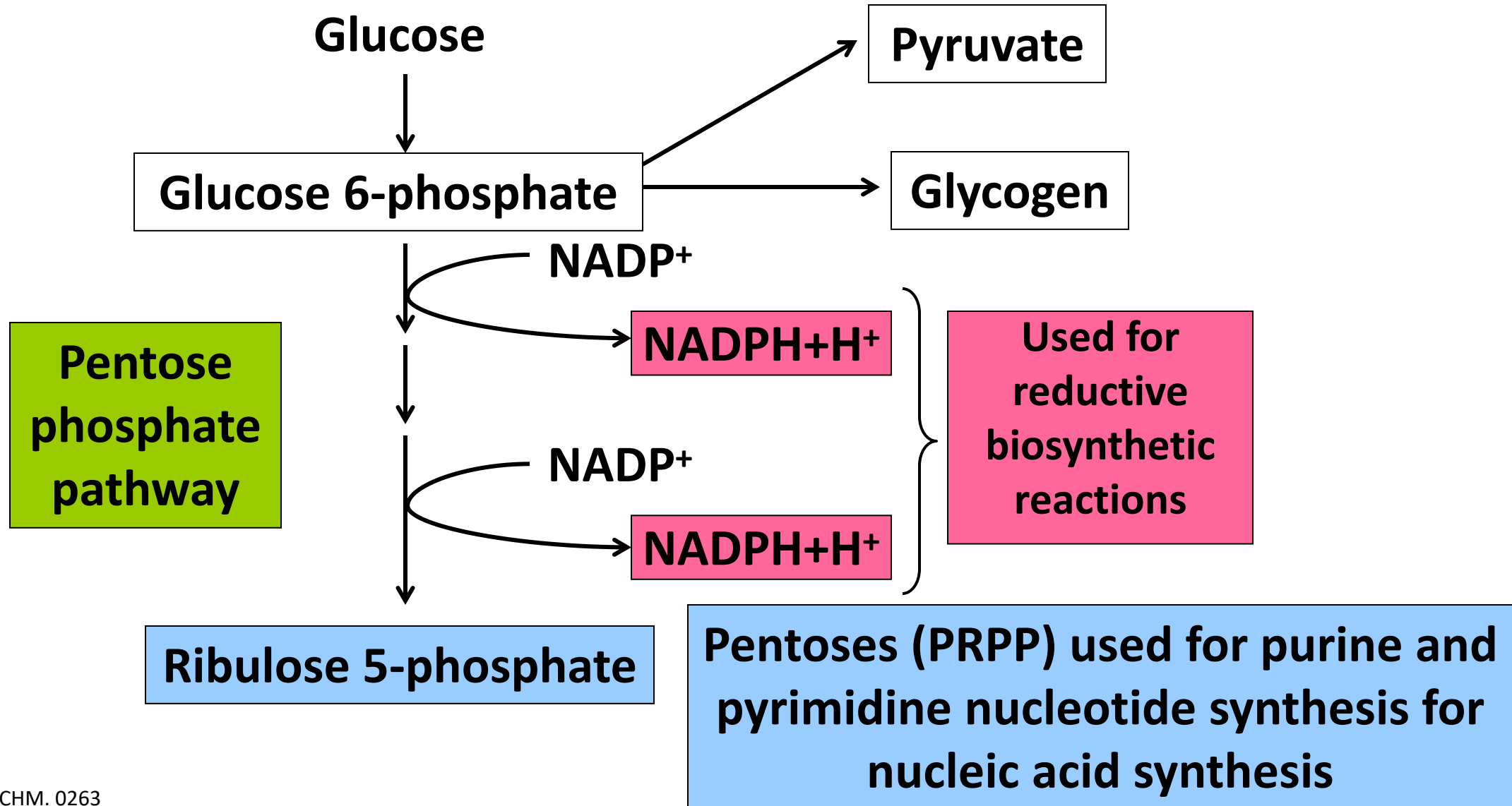
- [Pentose Phosphate Pathway and Nicotinamide Adenine Dinucleotide Phosphate | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- **DLA: Reactive oxygen species (After the lecture)**

Pentose phosphate pathway/ Hexose monophosphate shunt: RBCs and WBCs



- A.k.a. Hexose monophosphate shunt or 6-phosphogluconate pathway
- Occurs in **cytosol** of cells
- Alternate pathway for metabolism (oxidation) of glucose (pathway does **NOT** produce **energy/ATP**)
- Why do we need the pentose phosphate pathway???
 - Provides **NADPH + H⁺**
 - Provides **pentoses (PRPP)**: Purine and pyrimidine nucleotide biosynthesis

Overview of pentose phosphate pathway



Phases of pentose phosphate pathway

Oxidative phase

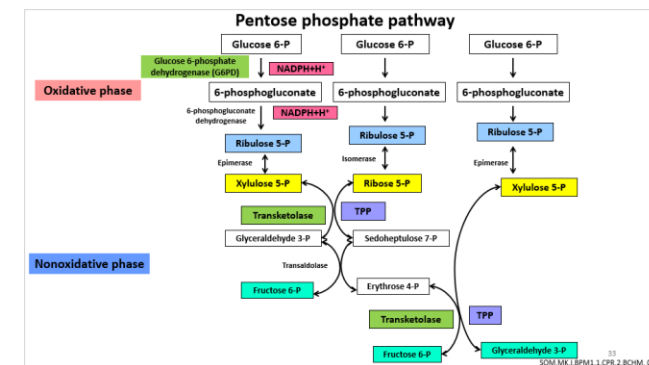
- Irreversible reactions
- Forms **NADPH** and **pentose phosphate**
- Enzymes: Dehydrogenases (**G6PD**)

Non-oxidative phase

- Reversible reactions
- Takes place under conditions where pentoses are not required by cells
- Involves **interconversion of sugars** resulting in formation of **glycolytic intermediates**
- Enzymes: **Transketolase (TPP)** and **transaldolase**

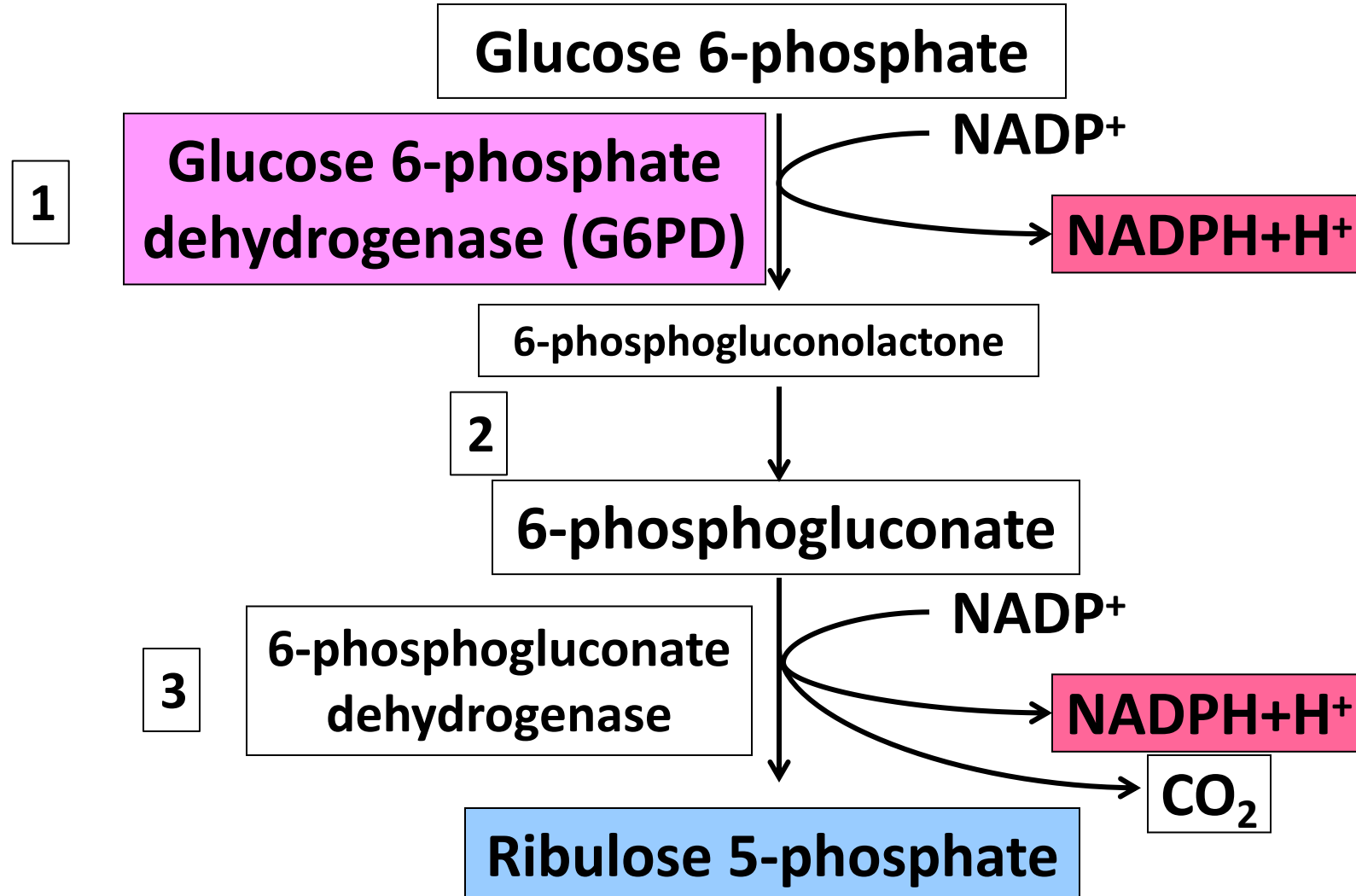
Pentose phosphate pathway

- Where is it active??
 - Liver, lactating mammary gland, adipose tissue (NADPH for fatty acid biosynthesis)
 - Adrenal cortex, testes, ovaries (NADPH for synthesis of steroid hormones)
 - **Erythrocytes (NADPH for reduction of glutathione)**
 - **White blood cells and macrophages (NADPH for phagocytosis)**



Oxidative phase of the pathway

Regulated
enzyme and
disorder
associated



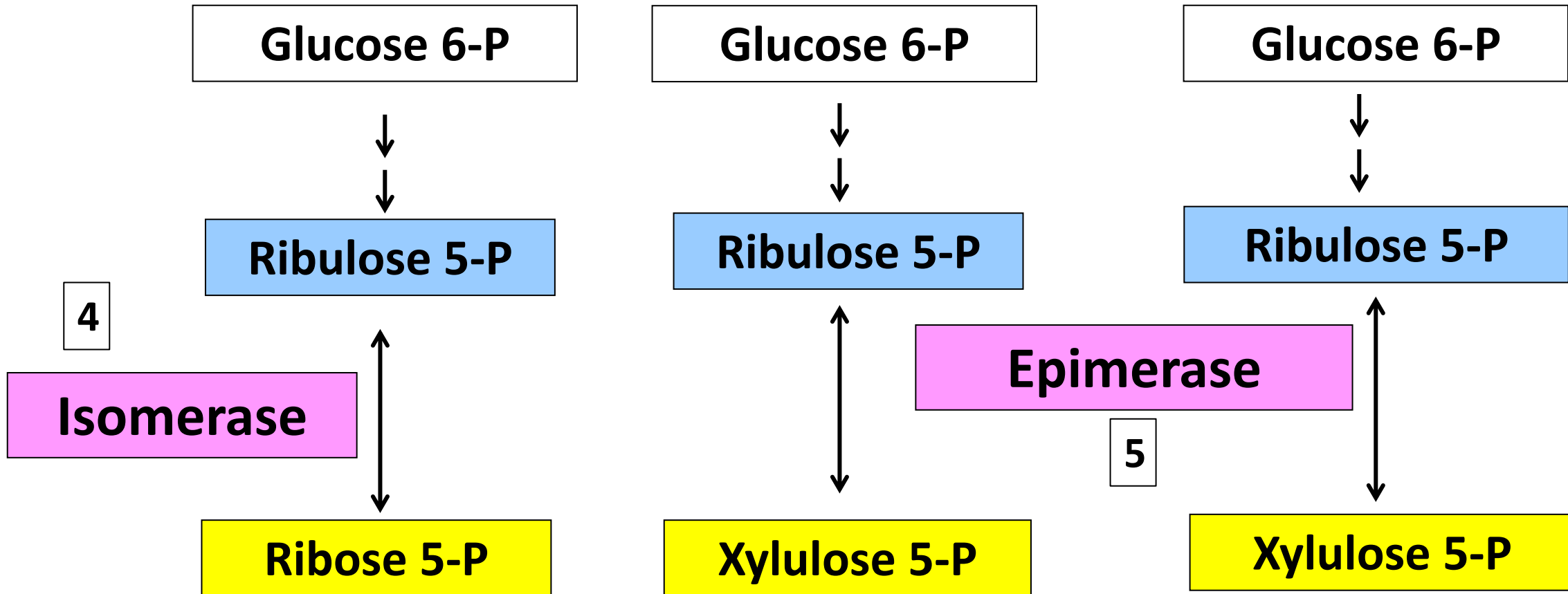
Oxidative phase of pathway

- Catalyzed by **G6PD (1)** and 6-phosphogluconate dehydrogenase (3)
- Both reactions form **NADPH+ H⁺** (reducing equivalents)
- Oxidative decarboxylation by 6-phosphogluconate dehydrogenase enzyme forms ribulose 5-phosphate (**pentose** phosphate)
- Regulatory enzyme of pathway: **G6PD**

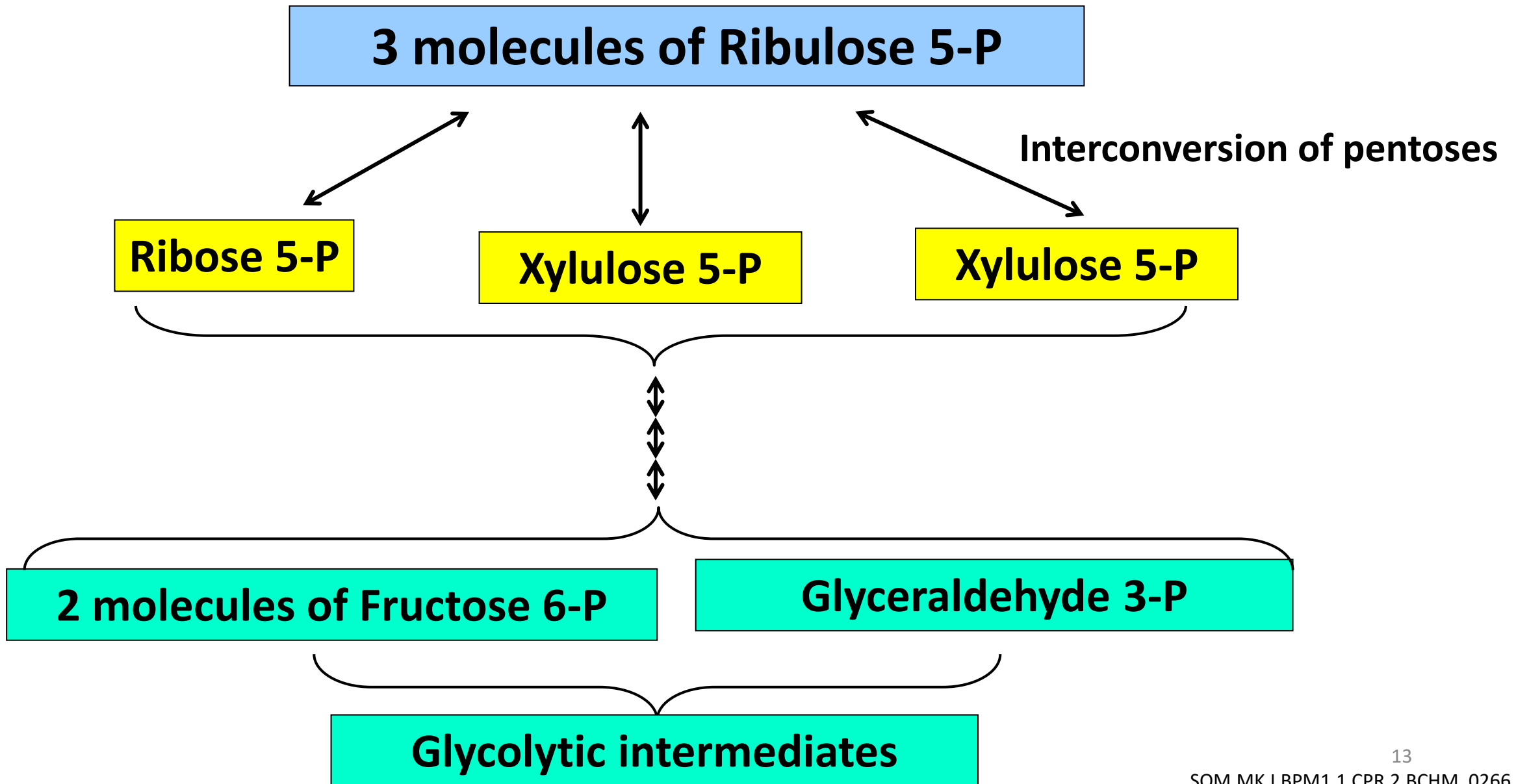
Non-oxidative phase of pentose phosphate pathway

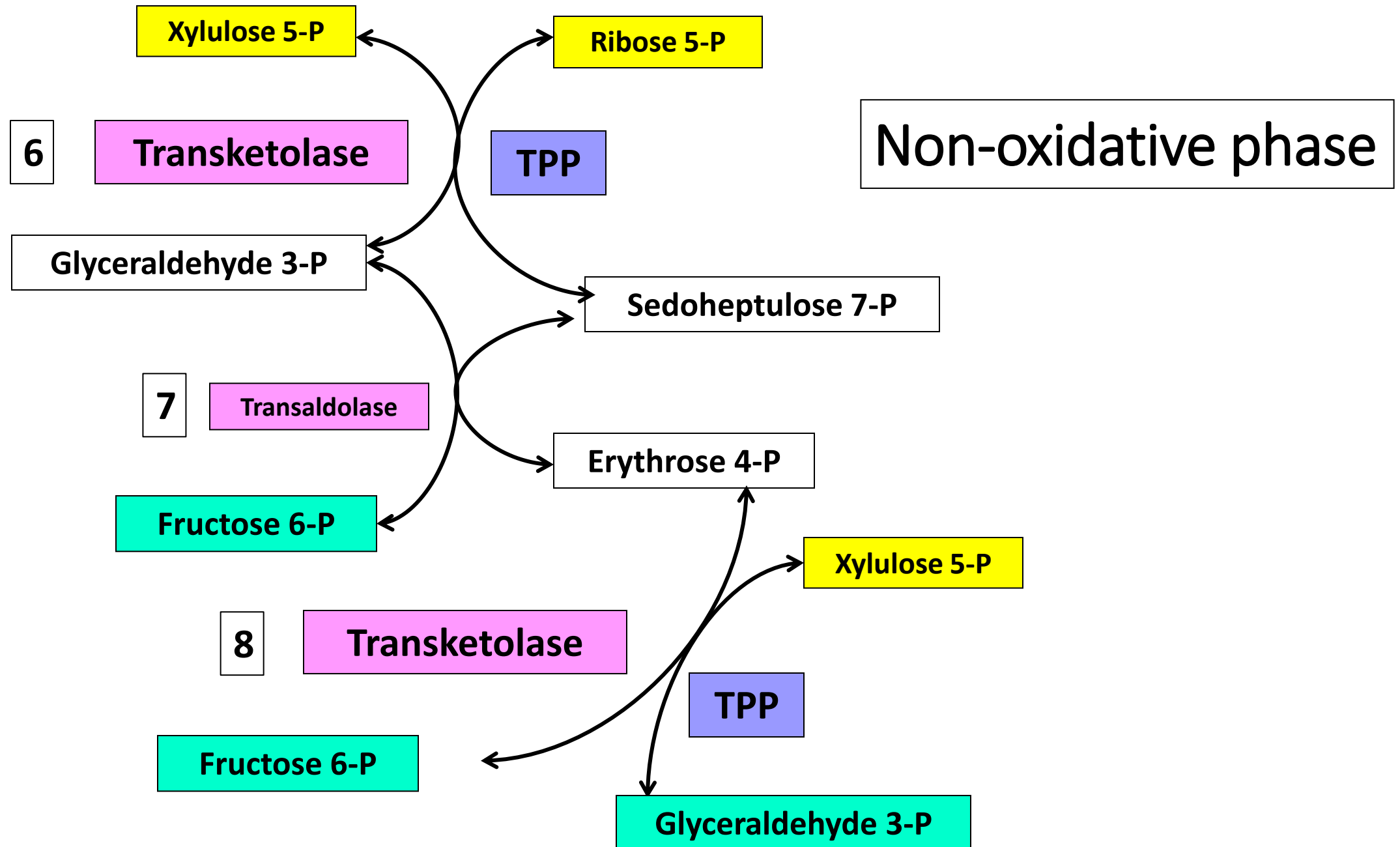
- Occurs in cells/ tissues that do not require pentose phosphate
- Convert excess ribulose 5-phosphate into **glycolytic intermediates** (excess pentoses are metabolized by major pathways) – to recycle carbon atoms
- To understand the non-oxidative phase, we imagine that 3 molecules of glucose 6-phosphate are converted to 3 molecules of ribulose 5-phosphate by oxidative reactions

Non-oxidative phase: Interconversion of pentoses

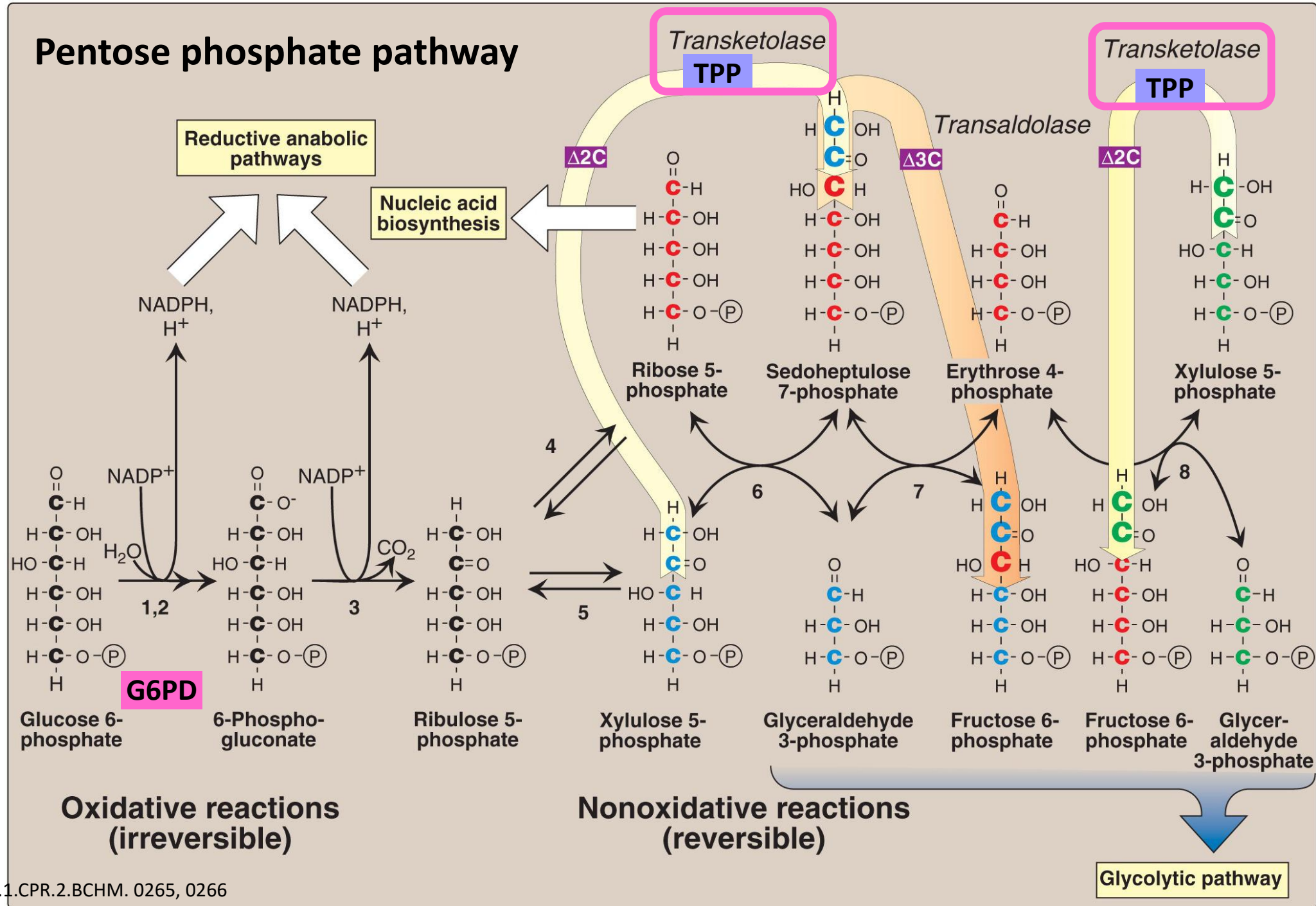


Overview of non-oxidative phase of PPP





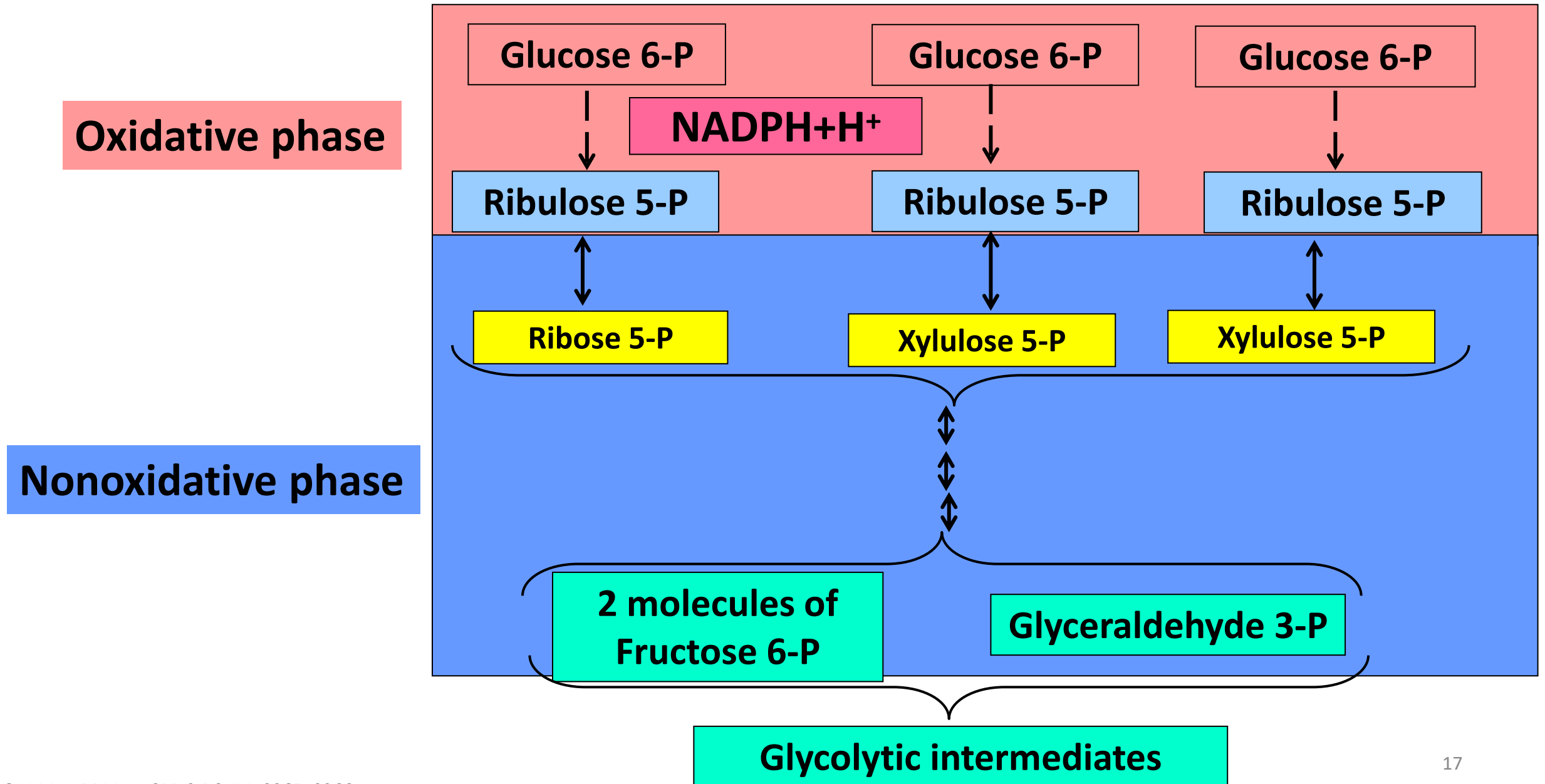
Pentose phosphate pathway



Transketolase and thiamine (TPP)

- Transketolase uses TPP (thiamine pyrophosphate) as coenzyme
- Transketolase activity in RBCs used an index of **nutritional thiamine (vitamin B1) status**
- Laboratory test to identify nutritional thiamine deficiency
- Patients with beriberi (thiamine deficiency) have low erythrocyte transketolase activity

Overview of pentose phosphate pathway



Pentose phosphate pathway depends on cell requirements

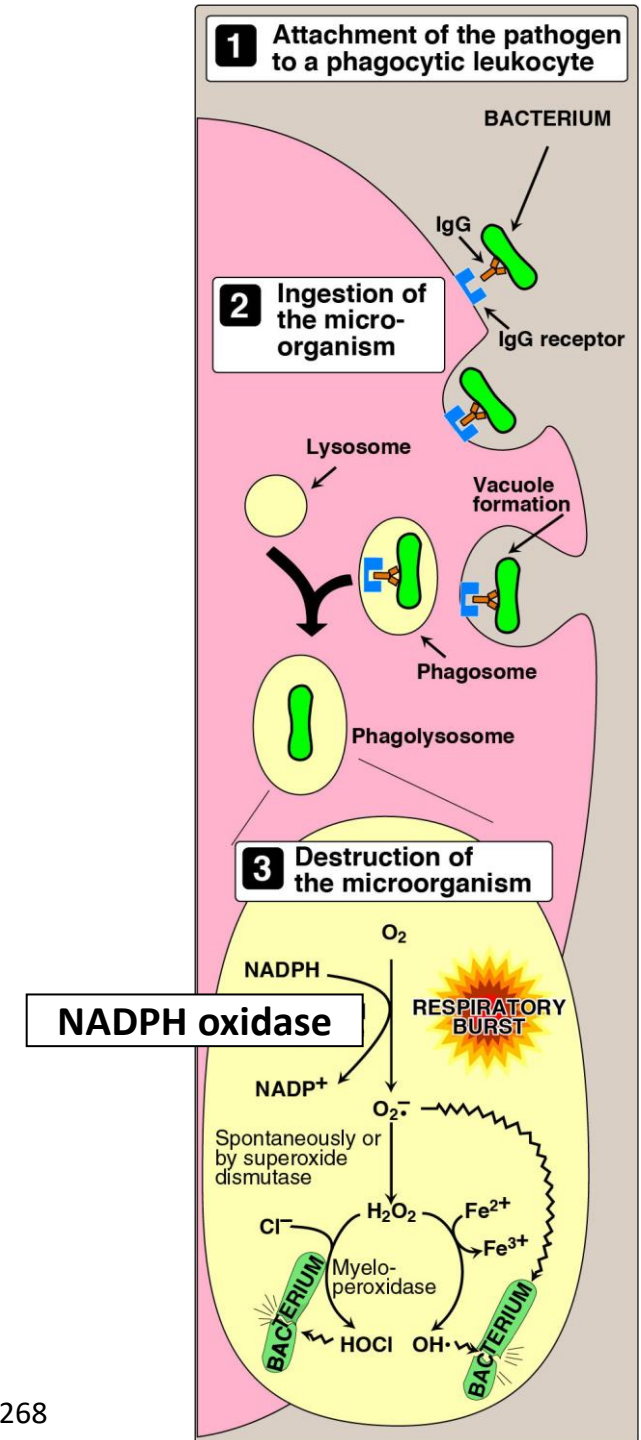
- If cells need both pentoses and NADPH, only the oxidative phase operates
- If cells need only NADPH (RBCs), Oxidative phase forms NADPH and the excess pentoses are recycled to glycolytic intermediates (via non-oxidative phase)
- If cells need only pentoses, then the glycolytic intermediates are used to synthesize the pentoses (non-oxidative phase in the reverse direction).

Uses of NADPH

- Reductive biosynthetic reactions in fatty acid synthesis and synthesis of steroid hormones/ cholesterol
- **RBCs: Detoxification of hydrogen peroxide (H_2O_2) and reactive oxygen species**
- Cytochrome P450 system (drug metabolizing enzymes)
- **Phagocytosis in WBCs (Forms reactive oxygen species)**
- Synthesis of nitric oxide (NO)

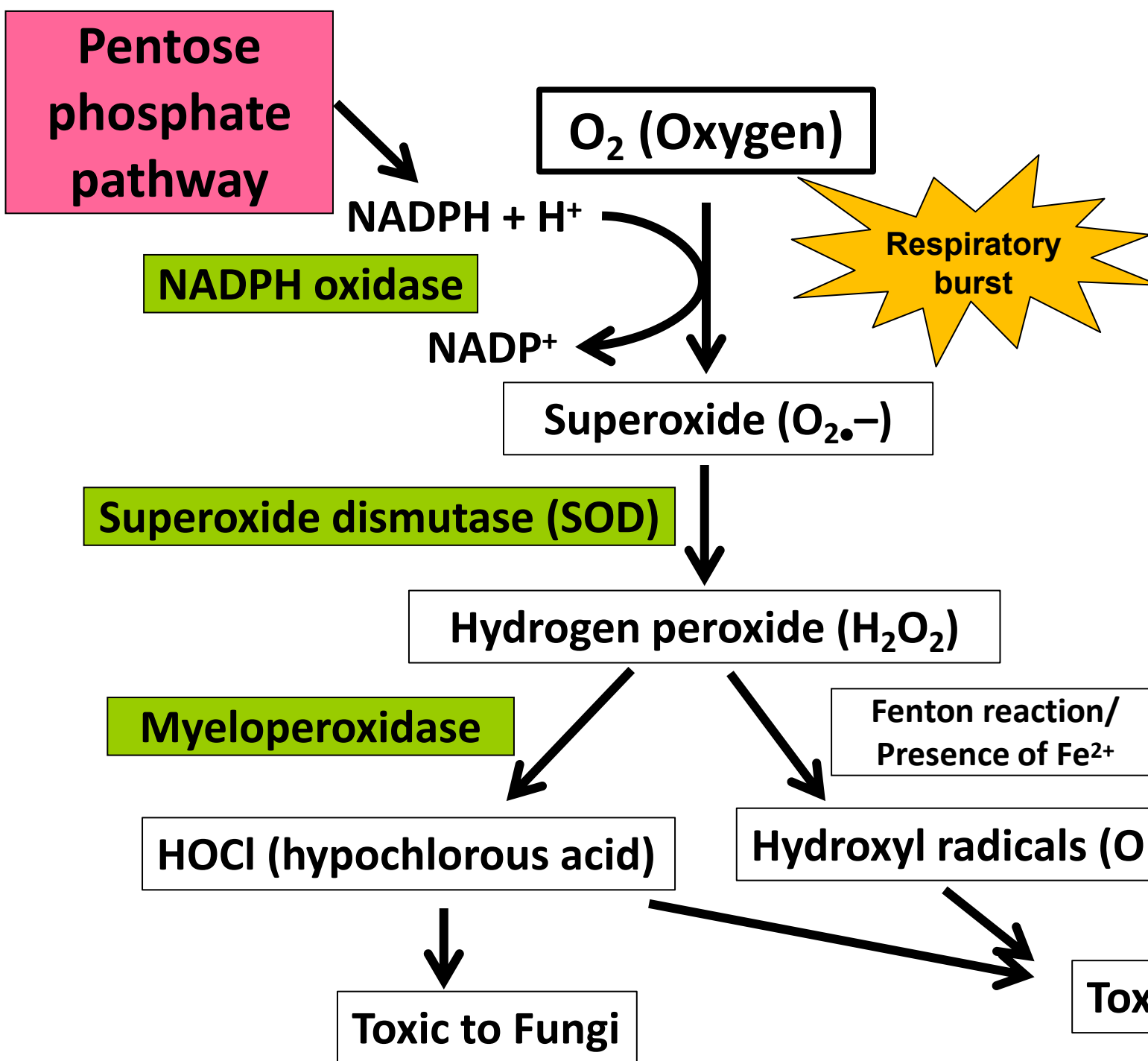
Phagocytosis by WBCs

- On phagocytosis, neutrophils and macrophages **increase their oxygen consumption** and increase generation of reactive oxygen species - '**Respiratory or oxidative**' burst
- **Superoxide free radicals** ($O_2^{\bullet-}$) formed using **NADPH oxidase** (requires NADPH)
- Superoxide $\rightarrow H_2O_2$ by **Superoxide dismutase (SOD)**
- H_2O_2 forms **hydroxyl radicals** ($OH^{\bullet}$) (toxic to bacteria)
- H_2O_2 forms **HOCl (hypochlorous acid)** kills bacteria and fungi
- **Infections result in increased production of oxidants**



Phagocytosis by WBCs (Macrophages)

NADPH oxidase activity responsible for respiratory burst (Increased oxygen consumption) following phagocytosis



SOM.MK.I.BPM1.1.CPR.2.BCHM. 0268, 0277

Inherited disorders characterized by defective phagocytosis

- **NADPH oxidase deficiency: Chronic granulomatous disease (CGD):** Severe persistent **bacterial** and fungal infections
 - Neutrophils show **lack of** increased oxygen consumption on phagocytosis (**defective respiratory burst/ negative respiratory burst test**)
- **Myeloperoxidase deficiency:** Increased risk of **fungal infections**
 - **Normal respiratory burst (Positive respiratory burst)** but decreased HOCl formation
- **G6PD deficiency usually does NOT affect WBCs**
- DLA: Reactive oxygen species
- Differentiate other immunodeficiency disorders: ADA and Di George syndrome

RBCs: Detoxification of H_2O_2 (Reactive oxygen species)

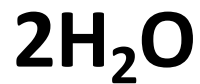
Drugs/ Infection/
Fava beans

↓ 1

Reactive oxygen species



Glutathione peroxidase
(Selenium)



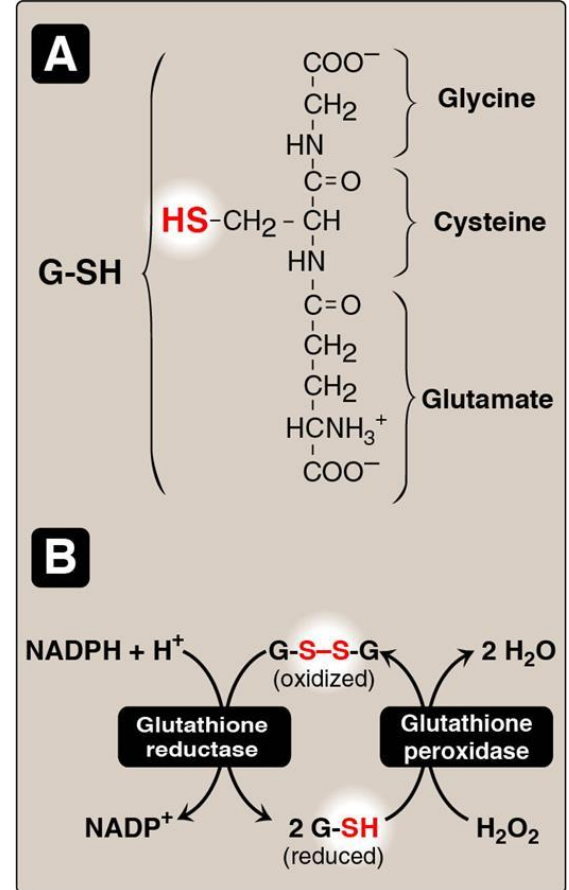
2 GSH
(reduced)

GS-SG
(oxidized)

NADP⁺

Glutathione reductase

NADPH+H⁺



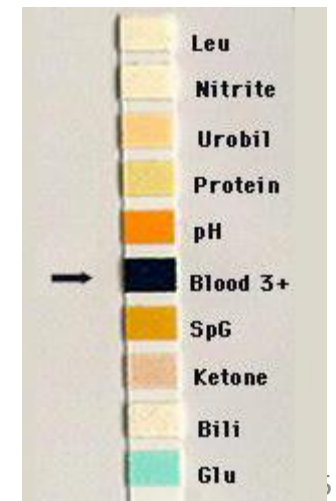
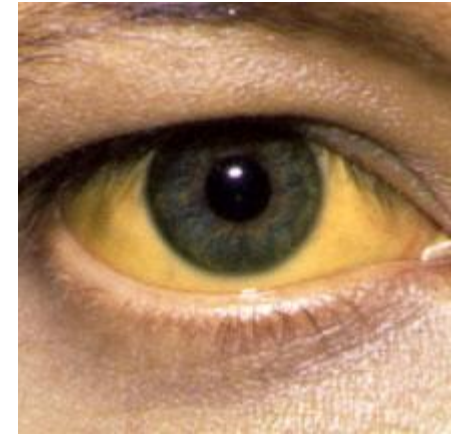
Pentose phosphate pathway

RBCs: Detoxification of H_2O_2 (Reactive oxygen species)

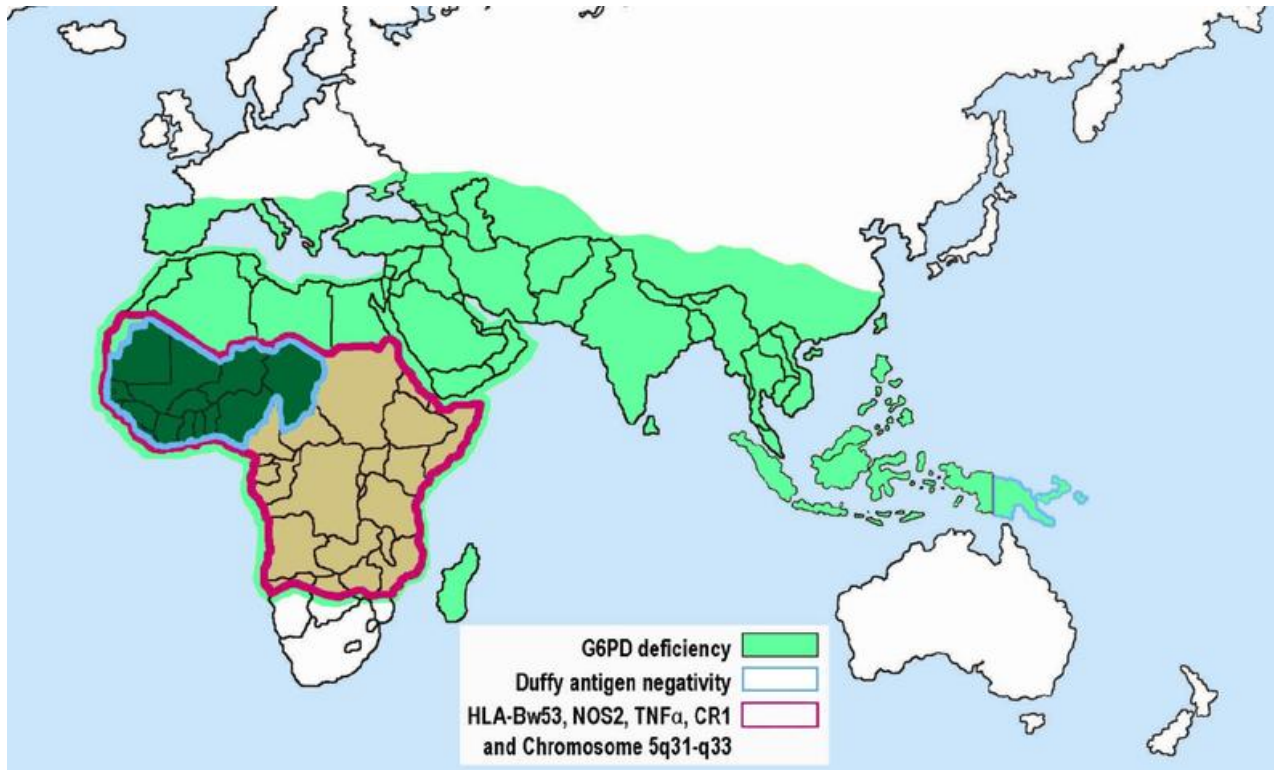
1. Oxidant drugs (Sulfa drugs/Primaquine) or infection or fava beans increase production of **free radicals (reactive oxygen species)**
 2. In RBCs, H_2O_2 detoxified to H_2O by **glutathione peroxidase**
 - Hydrogen donated by **reduced glutathione (GSH)**
 - **Selenium containing enzyme**
 3. Oxidized glutathione (GS-SG) converted to reduced glutathione (GSH) by **glutathione reductase**, using NADPH formed by pentose phosphate pathway
- GSH (Antioxidant) to detoxify H_2O_2 (ROS) in RBCs
 - Defect in pentose phosphate pathway results in H_2O_2 accumulation (reactive oxygen species) that damage RBC membrane resulting in hemolysis

G6PD deficiency: Case report

- A 16-year-old boy is diagnosed with pneumonia. He is treated with **sulfamethoxazole (sulfa drugs/sulfonamides)**.
- On the third day of treatment, he is slightly jaundiced (increased serum bilirubin), and he feels fatigued.
- His hemoglobin has fallen by 3.5g/dL (compared to previous value)
- Urine is reddish brown and is positive for hemoglobin
- Evidence of **acute hemolysis**

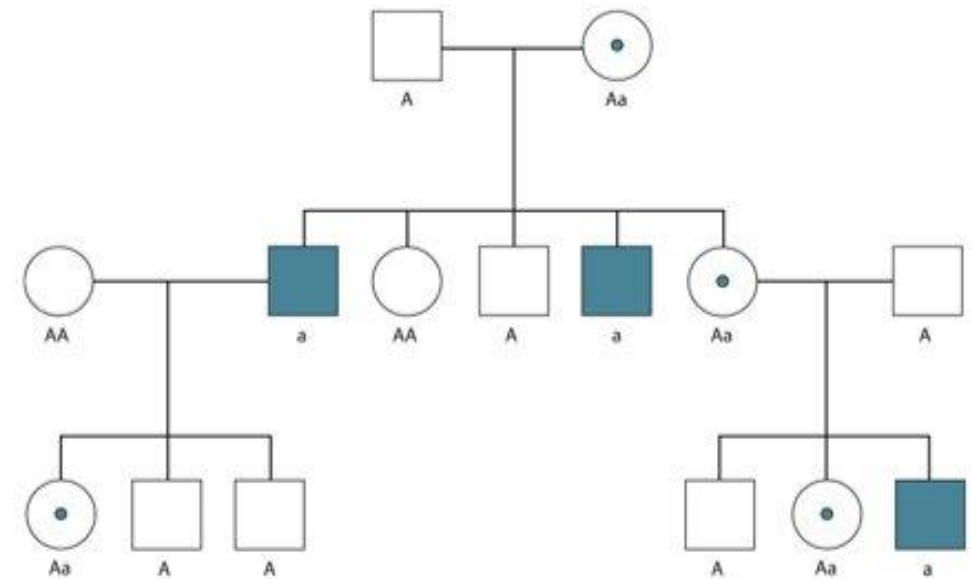


G6PD deficiency more common where malaria is endemic



G6PD is X-linked recessive

G6PD mutations (and many other mutations affecting RBC) might provide a selective advantage due to protection against the malarial parasite



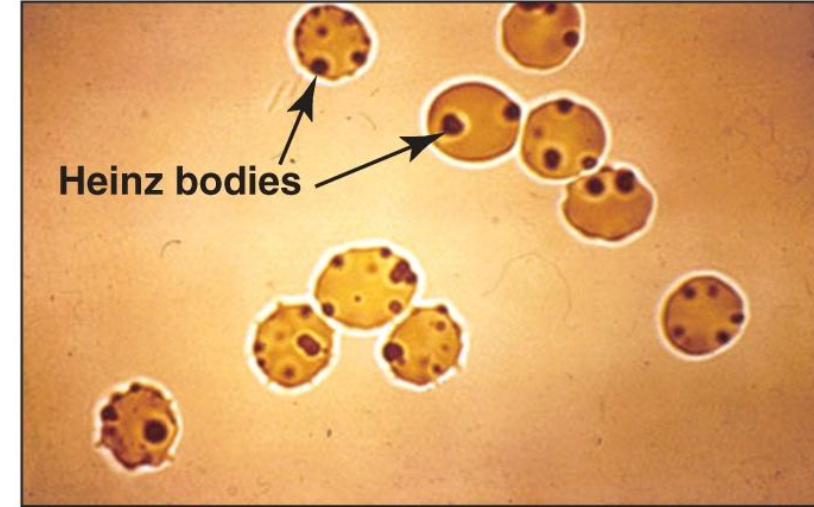
Human Genetics and Genomics, Fourth Edition. Bruce R. Korf and Mira B. Irons.
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G6PD deficiency

- Most common enzyme abnormality resulting in **hemolysis**
- **X-linked recessive disorder**
- Highest prevalence in Middle East, Africa, Asia and Mediterranean
- Reduced **G6PD** activity (pentose phosphate pathway) → Impairs NADPH formation
- Decreases cellular **reduced glutathione pool** (GSH main antioxidant in RBCs)
- Mature RBCs unable to synthesize new enzyme (do not have nucleus and cellular organelles)
 - **Older RBCs** more susceptible to hemolysis

G6PD deficiency: Effects on RBCs

- Impaired detoxification of free radicals and peroxides (Impaired GSH regeneration)
 - H_2O_2 accumulates → RBC membrane damage (unsaturated fatty acids in membrane phospholipids), resulting in **hemolysis**
- GSH maintains thiol (-SH) groups in proteins
 - G6PD deficiency results in formation of denatured insoluble proteins (**Heinz bodies**)
 - Due to inappropriate disulfide bond formation
- G6PD deficiency **most severe in RBCs**
 - PPP is ONLY pathway generating NADPH in RBCs
 - Mature RBCs unable to synthesize new enzyme



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Heinz bodies in RBCs

Mechanism of hemolysis in G6PD deficiency

Oxidative stress: Infections/ fava beans/ antimalarials/ sulfa drugs



GSH converted to GS-SG for detoxifying oxidants
(GSH used up for detoxification of oxidants)



Increased requirement of NADPH to convert oxidized glutathione to reduced form (GSH) –
(Remember NADPH production in G6PD deficiency is low)



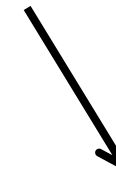
Oxidants accumulate



Damage of fatty acids containing double bonds (cell membrane phospholipids)



RBC lysis



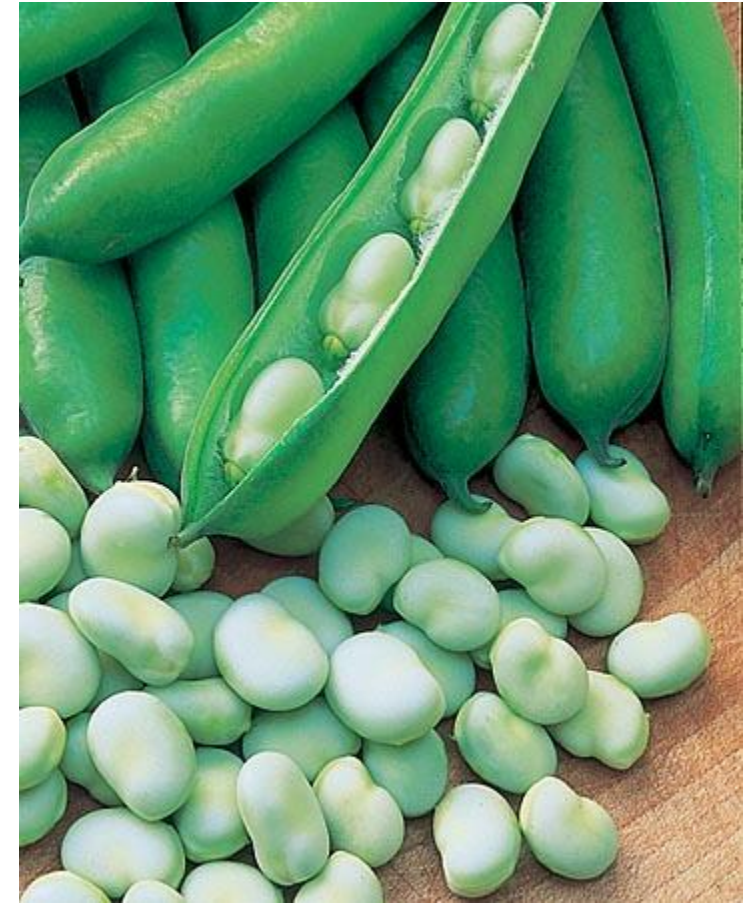
Denaturation of proteins due to GSH depletion



Formation of Heinz bodies
in RBC

G6PD deficiency

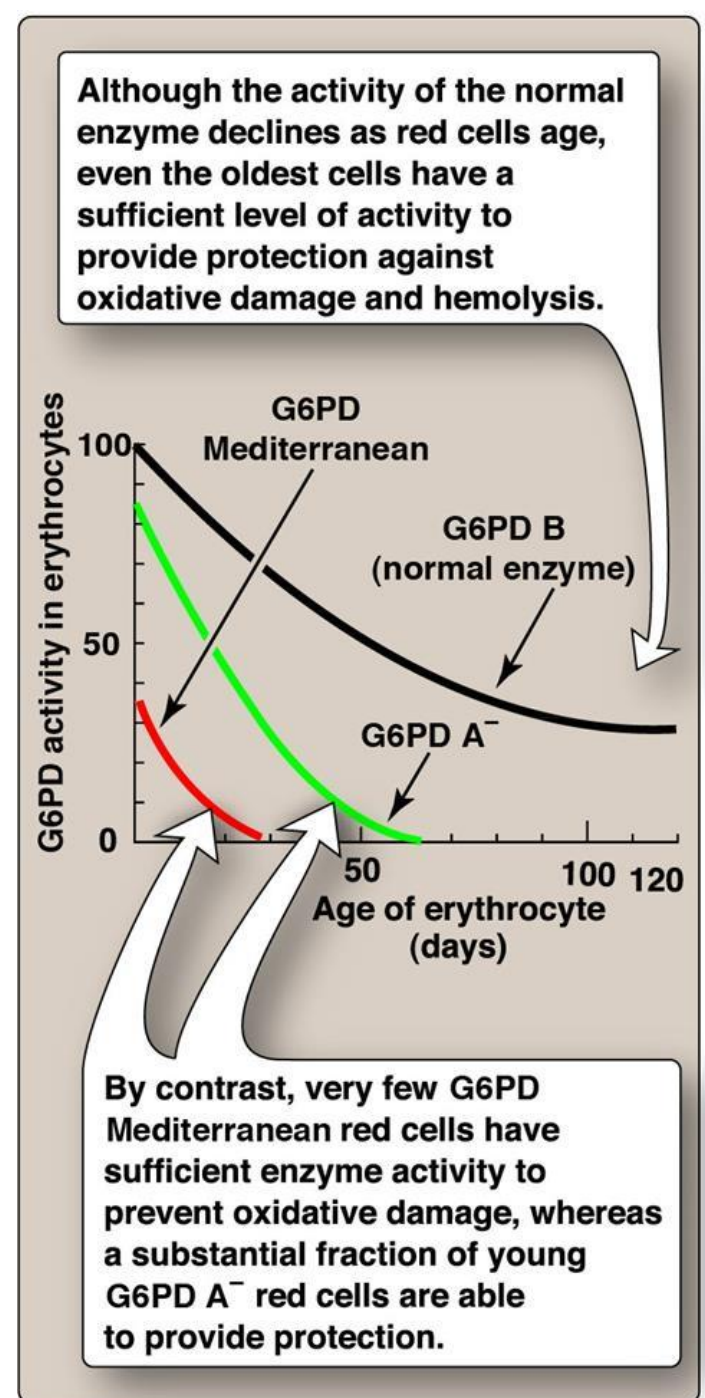
- Old (senile) RBCs vulnerable to G6PD deficiency, as mature RBCs have no nucleus or ribosomes, and cannot synthesize new enzyme
- **Precipitating factors (oxidants)**
 - Oxidant drugs
 - Antibiotics (Sulfa drugs)
 - Antimalarials (Primaquine)
 - Antipyretics (Aspirin)
 - Fava (broad) beans – ‘Favism’ (Mediterranean)
 - Infections



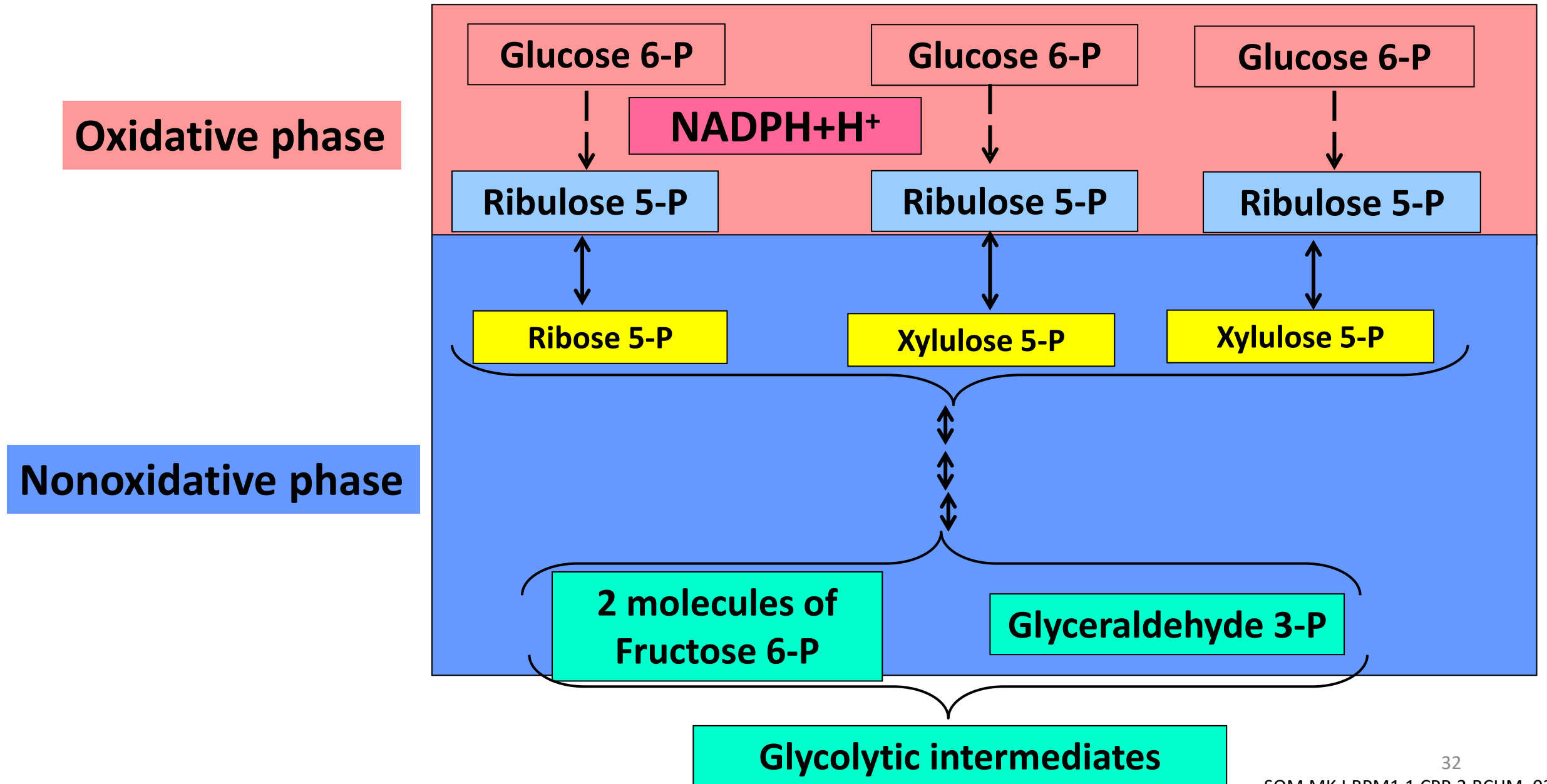
G6PD deficiency variants

- About 400 pathogenic variants of G6PD known
- G6PD A⁻ : Moderate hemolysis
- G6PD Mediterranean variant: Severe hemolysis

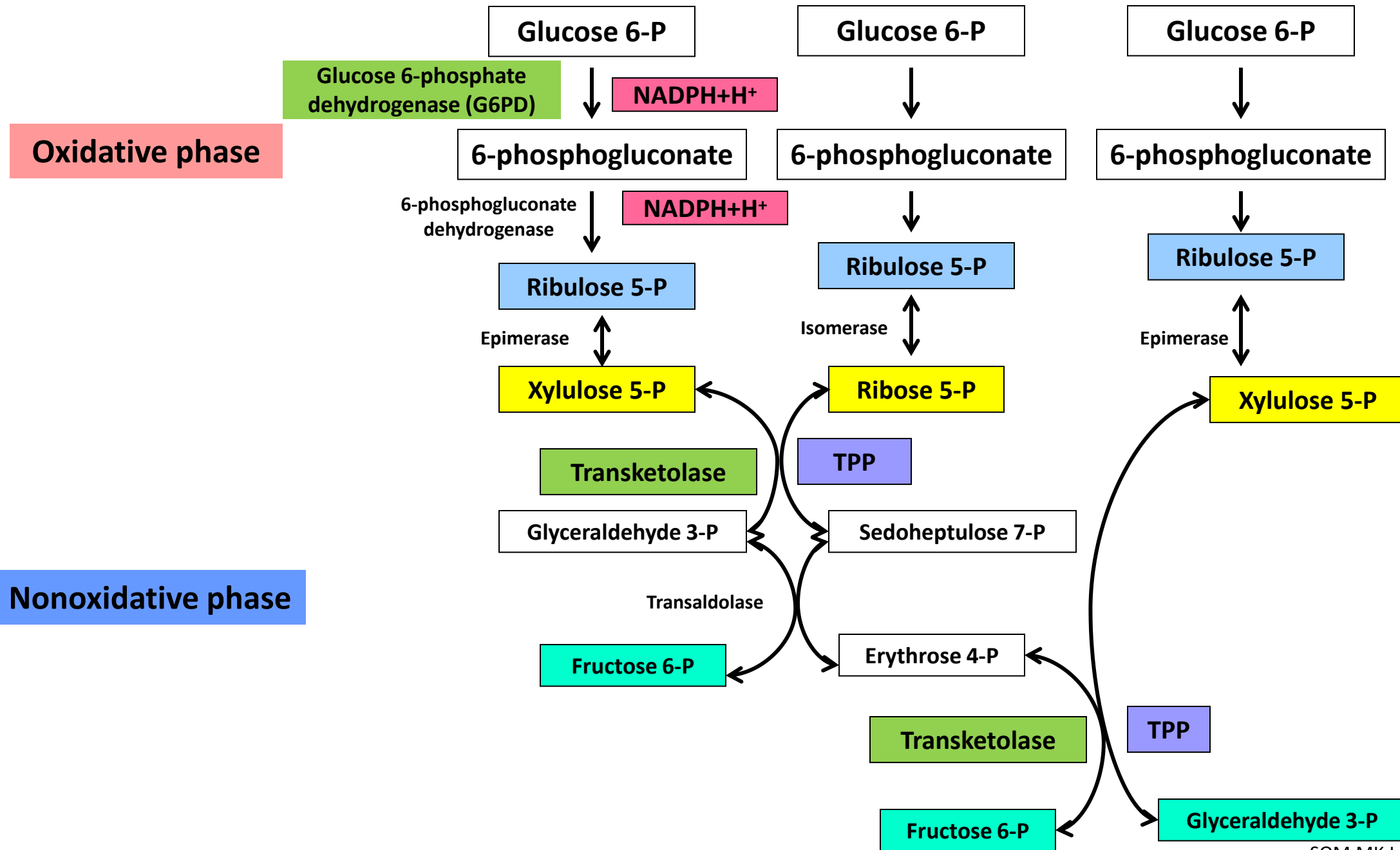
Allelic Heterogeneity



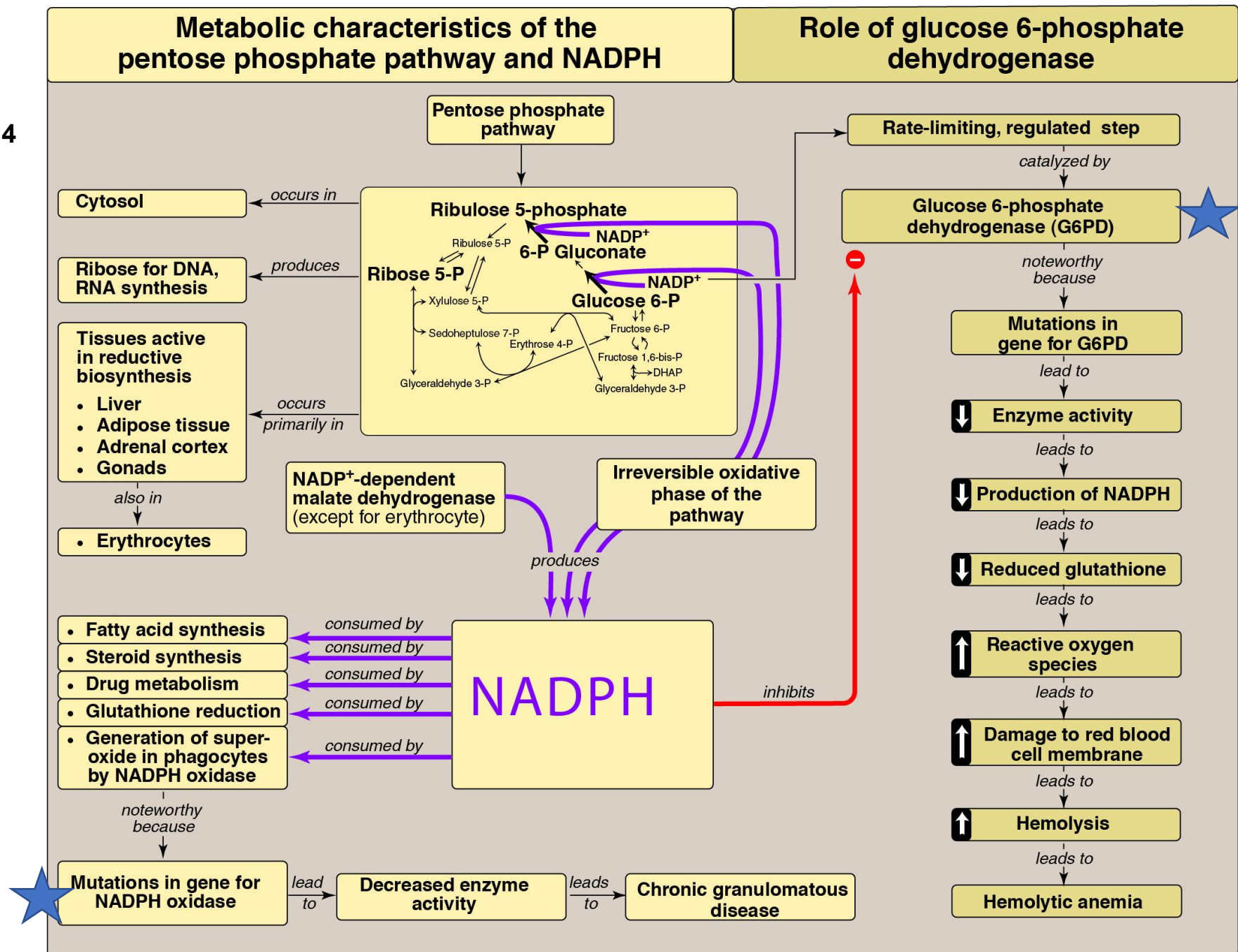
Overview of pentose phosphate pathway



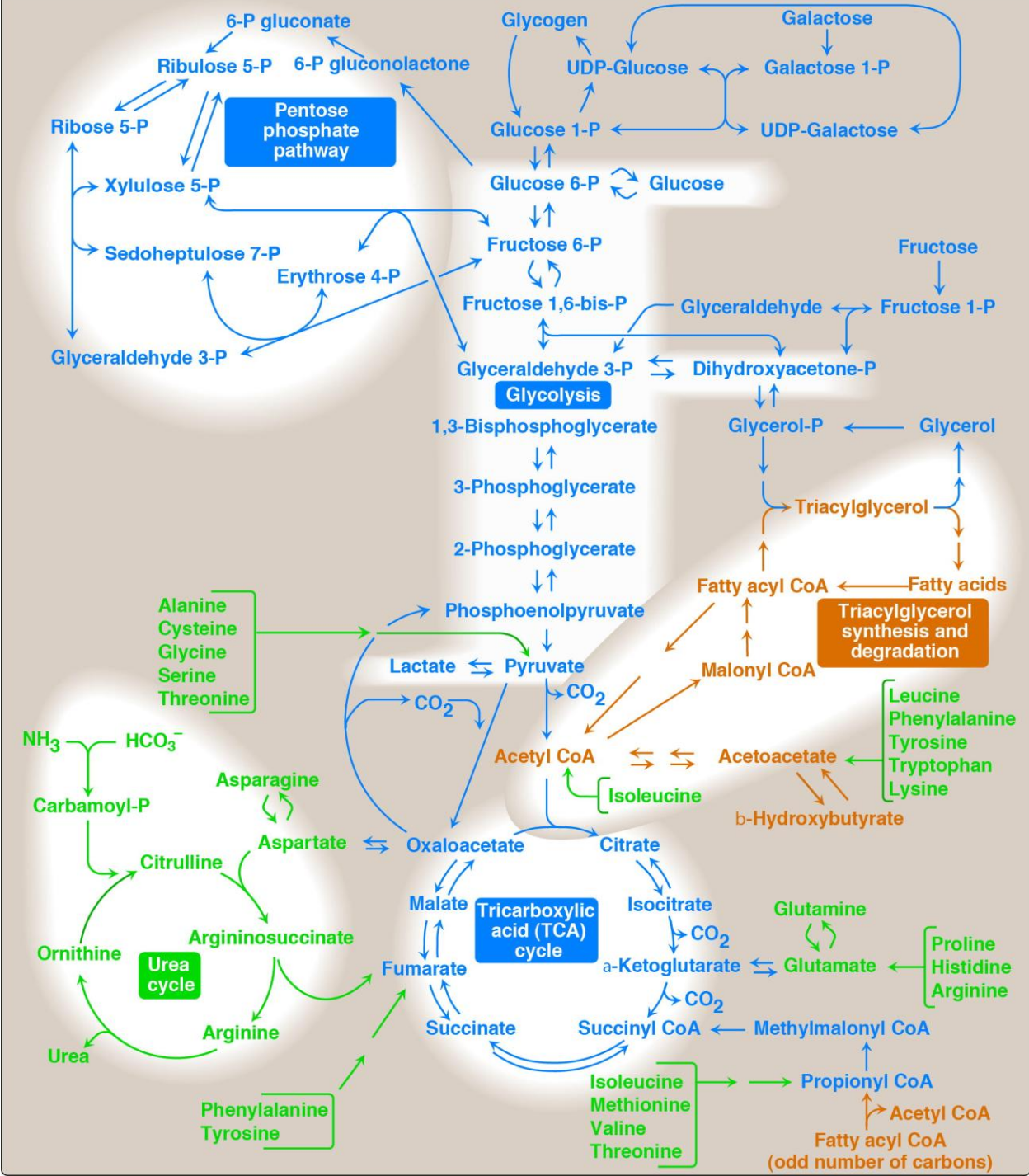
Pentose phosphate pathway



Concept map: Fig 13.14



Intermediary metabolism



Post-lecture required reading and objectives

- **DLA_Reactive oxygen species**

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0270	Define the terms radical, free radical and reactive oxygen species (ROS). Discuss cellular damage and effects of ROS.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0271	Describe formation of superoxide anion radical related to the ETC or cytochromes.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0272	Describe the reactions catalyzed by superoxide dismutase, catalase and glutathione peroxidase and indicate the ROS scavenged. Discuss the importance of selenium and indicate the enzyme containing selenium.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0273	Discuss the formation of hydroxyl radicals related to ionizing radiation, Haber- Weiss reaction and Fenton reaction.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0274	Discuss the compartmentation of free radical defense in the cell.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0275	Indicate non-enzymatic radical scavengers. Include vitamin C and E (found in fruits and vegetables).
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0276	Indicate the biochemical defect in amyotrophic lateral sclerosis (ALS).
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0277	Discuss the beneficial role of ROS formation as an immunological response in phagocytosis and describe the reactions catalyzed by NADPH oxidase and myeloperoxidase.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0278	Discuss the direct and potential effects of excess nitric oxide formation. Discuss the importance of inducible nitric oxide synthase for formation of RNOS.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0279	Predict the effect of NADPH oxidase deficiency and myeloperoxidase deficiency. Indicate the biochemical defect in chronic granulomatous disease.